D-GRIL: End-To-End Topological Learning with 2-Parameter Persistence

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Abstract

End-to-end topological learning using 1-parameter persistence is well-known. We show that the framework can be enhanced using 2-parameter persistence by adopting a recently introduced 2-parameter persistence based vectorization technique called Gril. We establish a theory for gradient *descent* on Gril producing D-Gril. We show that D-Gril can be used to learn a bifiltration function on benchmark graph datasets. Further, we exhibit that this framework can be applied in the context of bio-activity prediction in drug discovery.

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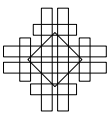
1 Introduction

In recent years, persistent homology, one of the flagship concepts of Topological Data Analysis (TDA), has found its use in many fields such as neuroscience, material science, sensor networks, shape recognition, gene expression data analysis and many more [12, 25]. The performance of machine learning models such as Graph Neural Networks (GNNs) can be improved by augmenting topological information captured by persistent homology [8, 15, 29, 30]. Classical persistent homology, also known as 1-parameter persistence, captures the evolution of topological structures in a simplicial complex K following a filter function $f\colon K \to \mathbb{R}$. The evolution of topological structures, in this case, can be completely characterized and compactly represented as *persistence diagrams* or equivalently *barcodes* [21, 50]. These

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persistence diagrams or barcodes can be vectorized [1, 2, 8, 28, 31, 42] and used in machine learning pipelines. In most applications, the simplicial complex K is given and the choice of the filter function f depends on the application. Choosing an appropriate filter function can be challenging. To avoid this, in [27], the authors proposed an end-to-end learning framework to learn the filter function rather than relying on making the right choice. They showed that learning the filter function performs better than the standard choices of filter functions on many graph datasets.

To obtain richer topological information, we can consider an $\mathbb{R}^n$-valued filter function instead of a scalar filter function. This leads to the structure of a multiparameter persistence module instead of 1-parameter persistence module. Multiparameter persistence modules, unlike their 1-parameter counterparts, can *not* be classified completely using a discrete invariant [5]. Consequently, vectorizing multiparameter persistence modules while retaining as much topological information as possible has become a challenging problem. There are many recent works in this direction which use incomplete invariants such as *rank invariant* [5] or equivalently *fibered barcodes* [33], multiparameter persistence images [6], multiparameter persistence landscapes [45], multiparameter persistence kernel [11], vectorization of signed barcodes [35], topological fingerprinting for virtual screening using multidimensional persistence [16], effective multidimensional persistence [10] or *generalized rank invariant* [48]. The authors, in all these works, show that 2-parameter persistence methods perform better than 1-parameter persistence methods on many graph and time-series datasets, suggesting that an $\mathbb{R}^2$-valued filter function, indeed, captures richer topological information than a scalar filter function. However, in all these works, one needs to make a suitable choice for an $\mathbb{R}^2$-valued filter function. Akin to the 1-parameter setup, learning the filter function rather than choosing a filter function can prove to be more informative and beneficial for the task at hand.

In this paper, we propose an end-to-end learning framework using 2-parameter persistent homology based on the vectorization Gril introduced by [48]. We show that Gril is piecewise affine and give an explicit formula for the differential of Gril. The results discussed in [7, 14] help us to show the convergence of stochastic sub-gradient descent on Gril. We use these results to build a differentiable topological layer D-Gril, consequently an end-to-end learning pipeline. We use D-Gril for bio-activity prediction and provide experimental results. One of the key advantages of using TDA in the context of bio-activity prediction is its ability to capture and analyze the shape and structural characteristics of molecules in a manner that traditional methods might overlook. Molecules possess intricate shapes and topological features that directly influence their interactions with biological targets. TDA can uncover these subtleties, allowing for a deeper understanding of structure-activity relationships. We also show that D-Gril can be used, more generally, for bifiltration learning on graphs and provide experimental results where we compare with existing multiparameter persistence methods on benchmark graph datasets.

In a recent work, authors of [44] present an end-to-end topological learning method by developing a general framework suitable for a class of invariants of multiparameter persistence. Our approach though specific to Gril is more direct, uses simpler arguments, and is faster for training.

2 Overview

Our main construct is Gril, introduced in [48] for vectorizing a 2-parameter persistence module. Gril is defined in the same spirit as persistence landscapes [2], introduced for 1-parameter persistence. The landscape uses the rank function over the interval decomposition known for 1-parameter persistence modules. To extend the concept to 2-parameter persistence

modules, one faces two main challenges: (i) in general, persistence modules beyond 1-parameter may not decompose into intervals, (ii) the usual rank function cannot be defined for intervals that are not rectangular. GRIL solves (i) by defining the landscape function over a fixed set of two-dimensional intervals called ℓ-*worms* instead of the support of the indecomposables of the modules, and solves (ii) by considering a generalized version of the rank function termed as *generalized rank* (Definition 3).

An interesting property of the generalized rank over intervals is that it monotonically decreases as the intervals get larger. Thus, given fixed parameters $k > 0$ and $\ell > 0$, the width d of an ℓ-worm can be increased until the generalized rank over the worm drops below k. The maximum value of the width thus obtained, $d_{\mathbf{p}}$, is the value of GRIL at the worm center point $\mathbf{p}$ for k and ℓ.

GRIL yields a vector representation of the 2-parameter persistence module formed from a given simplicial complex and a bifiltration function (formally defined in section 3). However, there is no provision to learn this bifiltration function. To learn a suitable bifiltration function, we build a differentiable layer called D-GRIL. We show in section 6 that the learnt bifiltration function performs better than some of the popular bifiltration functions known in TDA supporting the need for end-to-end learning.

3 Background

We review and recall some definitions and results in this section. Section 3.1 primarily consists of concepts in multiparameter persistent homology. Section 3.3 consists of concepts and results that establish the conditions required for the convergence of stochastic sub-gradient descent. Section 3.2 contains some background which is required to prove that GRIL satisfies these conditions.

3.1 Multiparameter persistent homology

In this subsection, we briefly review the concepts in multiparameter persistent homology. We focus, primarily, on 2-parameter persistent homology. For detailed definitions, refer [20, 22]. We begin by recalling the definition of a simplicial complex. Given a finite vertex set V, a simplicial complex $K = K(V)$ defined on this vertex set is a collection of subsets of V such that if a subset $\sigma \subseteq V$ is in K, then all proper subsets $\tau \subset \sigma$ are also in K. Each element in K with cardinality $k+1$ is called a k-simplex or simply a simplex. A graph is a simplicial complex where V is the vertex set of the graph and K consists of edges and vertices of the graph. A *filtration* is a collection of simplicial complexes $\{K_{\mathbf{x}}\}_{\mathbf{x}\in\mathbb{R}}$ indexed by the reals, with the property that $K_{\mathbf{x}} \subseteq K_{\mathbf{y}}$ for each $\mathbf{x} \leq \mathbf{y}$. Extending this definition to collections of complexes indexed by $\mathbb{R}^2$ with the product partial order ($\mathbf{x} \leq \mathbf{y} \in \mathbb{R}^2$ if $x_1 \leq y_1$ and $x_2 \leq y_2$), we get a *bifiltration*.

▶ **Definition 1** (Bifiltration). *A bifiltration is a collection of simplicial complexes* $\{K_{\mathbf{x}}\}_{\mathbf{x}\in\mathbb{R}^2}$ *where* $K_{\mathbf{x}} \subseteq K_{\mathbf{y}}$ *for all comparable* $\mathbf{x} \leq \mathbf{y} \in \mathbb{R}^2$.

Let K be a simplicial complex and $f\colon K \to \mathbb{R}^2$ be a map with the property that $f(\sigma) \leq f(\tau) \in \mathbb{R}^2$ for all $\sigma \subseteq \tau$. For each $\mathbf{x} \in \mathbb{R}^2$, let $K_{\mathbf{x}} \coloneqq \{\sigma \in K\colon f(\sigma) \leq \mathbf{x}\}$. Clearly, $K_{\mathbf{x}} \subseteq K_{\mathbf{y}}$ for all comparable $\mathbf{x} \leq \mathbf{y} \in \mathbb{R}^2$. The resulting bifiltration, denoted $\{K^f_{\mathbf{x}}\}_{\mathbf{x}\in\mathbb{R}^2}$, is called the *sub-level set* bifiltration of the *bifiltration function* f.

A filtration in the 1-parameter case induces a persistence module, obtained by considering the inclusion-induced linear maps between the vector spaces given by the homology groups of the simplicial complexes comprising the filtration. Similarly, we obtain a 2-parameter persistence module from a bifiltration.

▶ **Definition 2** (2-parameter persistence module). *A 2-parameter persistence module is an assignment of finite-dimensional vector spaces $M_{\mathbf{x}}$ for each $\mathbf{x} \in \mathbb{R}^2$, and of linear maps $M_{\mathbf{x}\leq\mathbf{y}} \colon M_{\mathbf{x}} \to M_{\mathbf{y}}$ for all comparable $\mathbf{x} \leq \mathbf{y} \in \mathbb{R}^2$, with the properties that $M_{\mathbf{x}\leq\mathbf{x}} = id$ and $M_{\mathbf{y}\leq\mathbf{z}} \circ M_{\mathbf{x}\leq\mathbf{y}} = M_{\mathbf{x}\leq\mathbf{z}}$ for all $\mathbf{x} \leq \mathbf{y} \leq \mathbf{z} \in \mathbb{R}^2$.*

Readers familiar with category theory can recognize that a 2-parameter persistence module is a functor $M \colon \mathbb{R}^2 \to \mathbf{vec}_{\mathbb{F}}$ where the poset $\mathbb{R}^2$ is regarded as a category and $\mathbf{vec}_{\mathbb{F}}$ denotes the category of finite-dimensional vector spaces over a field $\mathbb{F}$.

Given a bifiltration $\{K_{\mathbf{x}}\}_{\mathbf{x}\in\mathbb{R}^2}$, we consider $H_i(K_{\mathbf{x}})$, the ith homology group of the simplicial complex $K_{\mathbf{x}}$ over a field F, say $\mathbb{Z}_2$. Then, for each inclusion $K_{\mathbf{x}} \subseteq K_{\mathbf{y}}$, we get an induced linear map $H_i(K_{\mathbf{x}}) \to H_i(K_{\mathbf{y}})$. By this assignment, we get a 2-parameter persistence module.

Let $\{K^f_{\mathbf{x}}\}_{\mathbf{x}\in\mathbb{R}^2}$ be a sub-level set bifiltration. Then the 2-parameter persistence module M obtained by the homology assignment is called the 2-parameter persistence module induced by f, and we denote it as M_f.

As mentioned in Section 2, GRIL vectorization uses the concept of generalized rank introduced by [32], which we define below formally.

▶ **Definition 3** (Generalized Rank). *Let M be a 2-parameter persistence module. The restriction of M to an interval I of $\mathbb{R}^2$, denoted by $M|_I$, is the diagram formed by the collection of vector spaces $M_{\mathbf{x}}$ for $\mathbf{x} \in I$ and linear maps $M_{\mathbf{x}\leq\mathbf{y}}$ for all comparable $\mathbf{x} \leq \mathbf{y} \in I$. Then, the generalized rank of M over I is defined as the rank of the canonical map from the limit $\varprojlim M|_I$ to colimit $\varinjlim M|_I$:*

$$\mathsf{rk}^M(I) \coloneqq \mathit{rank}\left(\varprojlim M|_I \to \varinjlim M|_I\right).$$

We refer the reader to [36] for the definitions of limit, colimit, and the construction of the canonical limit-to-colimit map. Intuitively, generalized rank captures the number of independent topological features supported over the interval I. It can be computed efficiently for 2-parameter persistence modules [18] using zigzag persistence [17].

▶ Remark 4. In the special case where I is a rectangle, $\mathsf{rk}^M(I)$ is the rank of the linear map $M_{\mathbf{u}\leq\mathbf{v}}$ from the lower left corner $\mathbf{u}$ of the rectangle to the upper right corner $\mathbf{v}$.

3.2 o-minimal geometry

We recall the definitions of o-minimal structure and definable function here.

▶ **Definition 5** (o-minimal structure). *An o-minimal structure on $\mathbb{R}$ is a collection $\{S_n\}_{n\in\mathbb{N}}$ where each S_n is a set of subsets of $\mathbb{R}^n$ such that:*

1. *S_1 is exactly the collection of finite union of points and intervals.*
2. *S_n contains all the sets of the form $\{\mathbf{x} \in \mathbb{R}^n \colon p(\mathbf{x}) = 0\}$ where p is a polynomial on $\mathbb{R}^n$.*
3. *S_n is a Boolean sub-algebra of $\mathbb{R}^n$ for all n.*
4. *If $A \in S_n$ and $B \in S_m$, then $A \times B \in S_{n+m}$.*
5. *If $\pi \colon \mathbb{R}^{n+1} \to \mathbb{R}^n$ is the canonical projection onto the first n-coordinates, then for $A \in S_{n+1}$, $\pi(A) \in S_n$.*

A subset $A \in S_n$ for some $n \in \mathbb{N}$ is known as a *definable set* in the o-minimal structure. Let $A \in S_n$ be given. A function $f \colon A \to \mathbb{R}^m$ is said to be *definable* if the graph of the function in $\mathbb{R}^{n+m}$ is a definable set.

We note that all semi-algebraic functions are definable. In fact, the author in, [47], shows that there exists an o-minimal structure that simultaneously contains all semi-algebraic sets and the graph of the exponential function. The authors in [14] use this result to show that a

neural network, being a composition of definable functions, is definable (Corollary 5.11 [14]). For readers not familiar with the notion of definable functions, it is sufficient to know that a piecewise affine map is definable.

3.3 Stochastic sub-gradient descent

In this subsection, we briefly review the result and the conditions under which stochastic sub-gradient descent converges as shown in [14]. In [7], authors use this result in the setting of 1-parameter persistent homology and show that stochastic sub-gradient descent converges in that case. We begin by reviewing the definition of Clarke subdifferential. Given a locally Lipschitz function $f\colon \mathbb{R}^n \to \mathbb{R}$, the Clarke subdifferential of f at a point x is defined as:

$$\partial f(x) \coloneqq \mathbf{Co}\left\{s \in \mathbb{R}^n \colon \exists x^k \to x, \nabla f(x^k) \text{ exists, and } \nabla f(x^k) \to s\right\},$$

where **Co** denotes the convex hull of the set.

Now, given a loss function $\mathcal{L}$ with the objective of minimizing it, consider the differential inclusion

$$\dot{\mathbf{z}}(t) \in -\partial\mathcal{L}(\mathbf{z}(t)) \quad \text{for almost every } t,$$

where $\partial\mathcal{L}(\mathbf{z}(t))$ denotes the Clarke subdifferential of the function $\mathcal{L}$ at the point $\mathbf{z}(t)$. The solutions $\mathbf{z}(t)$ are the trajectories of the sub-gradient of $\mathcal{L}$ which can be approximated by the standard stochastic sub-gradient descent method given by:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k(\mathbf{y}_k + \zeta_k), \quad \mathbf{y}_k \in \partial\mathcal{L}(\mathbf{x}_k), \tag{1}$$

where the sequence $(\alpha_k)_k$ is the learning rate and $(\zeta_k)_k$ is a sequence of random variables or "noise". [14] establish that under mild technical assumptions (stated below) on these two sequences, the stochastic sub-gradient method converges almost surely to critical points of $\mathcal{L}$ if $\mathcal{L}$ is locally Lipschitz and definable.

Consider the following assumptions (Assumption C in [14]):

1. for any k, $\alpha_k \geq 0$, $\Sigma_{k=1}^{\infty} = \infty$ and $\Sigma_{k=1}^{\infty}\alpha_k^2 < \infty$;
2. $\sup_k ||x_k|| < \infty$, almost surely;
3. denoting by $\mathcal{F}_k$ the increasing sequence of σ-algebras $\mathcal{F}_k = \sigma(x_j, y_j, \zeta_j, j < k)$, there exists a function $p\colon \mathbb{R}^d \to \mathbb{R}$ which is bounded on bounded sets such that almost surely, for any k,

$$\mathbb{E}\left[\zeta_k|\mathcal{F}_k\right] = 0 \text{ and } \mathbb{E}\left[||\zeta_k||^2|\mathcal{F}_k\right] < p(x_k).$$

▶ **Proposition 6** (Corollary 5.9 [14]). *Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be a locally Lipschitz definable function*[1]*. Consider the iterates $\{\mathbf{x}_k\}_{k\geq 1}$ produced by the stochastic sub-gradient method (Eq (1)) and suppose Assumption C of [14] holds. Then, almost surely, every limit point of the iterates $\{\mathbf{x}_k\}_{k\geq 1}$ is critical for f and the function values $\{f(\mathbf{x}_k)\}_{k\geq 1}$ converge.*

4 Differentiability of GRIL

We begin this section by proving that Gril is a piecewise affine map in section 4.1. Gril being piecewise affine leads to stochastic sub-gradient converging almost surely on Gril, discussed in section 4.2. In section 4.3, we give an explicit formula for differential of Gril.

[1] The authors in [14] show that the result holds for a more general class of C^d-stratifiable functions.

4.1 GRIL as a piecewise affine map

We begin by recalling the definitions of ℓ-worm and GRIL.

▶ **Definition 7** (discrete ℓ-worm, [48]). *Let* $\boxed{\mathbf{p}}_d \coloneqq \{\mathbf{w} : \|\mathbf{p} - \mathbf{w}\|_\infty \leq d\}$ *be the* d*-square centered at* $\mathbf{p} \in \mathbb{R}^2$ *with side* $2d$*. Given* $\ell \geq 1$ *and* $d > 0$*, the* ℓ-worm, $\boxed{\mathbf{p}}_d^\ell$*, is defined as the union of all* d*-squares* $\boxed{\mathbf{q}}_d$ *centered at some point* $\mathbf{q}$ *on the off-diagonal line segment* $\mathbf{p} \pm \alpha \cdot (1, -1)$ *with* $\alpha = j \cdot d$ *where* $j \in \{1, \ldots, \ell - 1\}$.

Refer to Figure 1 for a 2-worm.

▶ **Definition 8** (GRIL, [48]). *For a 2-parameter persistence module* M*, the Generalized Rank Invariant Landscape (GRIL) is a function* $\lambda^M : \mathbb{R}^2 \times \mathbb{N}_+ \times \mathbb{N}_+ \to \mathbb{R}$ *defined as*

$$\lambda^M(\mathbf{p}, k, \ell) \coloneqq \sup\left\{d \geq 0 \colon \mathsf{rk}^M\left(\boxed{\mathbf{p}}_d^\ell\right) \geq k\right\}.$$

Fixing the bifiltration function f and k, ℓ, the landscape λ^M provides a function $\lambda^{M_f}_{k,\ell} : \mathbb{R}^2 \to \mathbb{R}$, $\mathbf{p} \mapsto \lambda^{M_f}(\mathbf{p}, k, \ell)$. The GRIL vector is the vector of values of $\lambda^{M_f}_{k,\ell}$ evaluated at a set of chosen sample points $\{\mathbf{p}_1, \ldots, \mathbf{p}_s\} \subset \mathbb{R}^2$.

Let K be a simplicial complex with n simplices, labelled $\sigma_1, \sigma_2, \ldots, \sigma_n$. A bifiltration function $f \colon K \to \mathbb{R}^2$ can be viewed as a vector $\mathbf{v}_f \in \mathbb{R}^{2n}$ where, for $k = 1, \ldots, n$, $\mathbf{v}_f[2k-1] = f_x(\sigma_k)$ and $\mathbf{v}_f[2k] = f_y(\sigma_k)$. Here, $f_x(\sigma)$ and $f_y(\sigma)$ denote the x- and y-coordinates of the vector $f(\sigma)$ respectively. Notice that the vectors in $\mathbb{R}^{2n}$ that correspond to a valid bifiltration function form a convex cone. We work with this set of vectors in $\mathbb{R}^{2n}$. In this setting, the authors in [48] show that GRIL is Lipschitz continuous in the following sense.

▶ **Proposition 9.** *Let* X *be a discrete space with* $|X| = n$*. For fixed* $k, \ell, \mathbf{p}$*, let* $\Lambda^{\mathbf{p}}_{k,\ell} \colon \mathbb{R}^{2n} \to \mathbb{R}$ *be the map* $\mathbf{v}_f \mapsto \lambda^{M_f}_{k,\ell}(\mathbf{p})$*. Then,* $\Lambda^{\mathbf{p}}_{k,\ell}$ *is Lipschitz continuous.*

By Rademacher's theorem [23], $\Lambda^{\mathbf{p}}_{k,\ell}$ is differentiable almost everywhere.

Let $\mathcal{G}_{k,\ell} \colon \mathbb{R}^{2n} \to \mathbb{R}^s$ be the map defined as:

$$\mathcal{G}_{k,\ell}(v_f) = \left[\Lambda^{\mathbf{p}_1}_{k,\ell}(\mathbf{v}_f), \Lambda^{\mathbf{p}_2}_{k,\ell}(\mathbf{v}_f), \ldots, \Lambda^{\mathbf{p}_s}_{k,\ell}(\mathbf{v}_f)\right]^T \tag{2}$$

where $\{\mathbf{p}_j\}_{j=1}^s$ are the s sampled center points. We drop the k, ℓ and refer to $\mathcal{G}_{k,\ell}$ as $\mathcal{G}$ whenever k, ℓ are well understood. We note that $\mathcal{G}$ is also a function of the center points $\mathbf{p}_j$ for all j. We show that $\mathcal{G}$ is piecewise affine.

For notational convenience, in what follows we denote $f_x(\sigma)$ as σ^x and $f_y(\sigma)$ as σ^y, and we call them the *simplex coordinates* of σ. Similarly, we denote the x-coordinate and y-coordinate of $\mathbf{p}$ as $\mathbf{p}^x$ and $\mathbf{p}^y$ respectively.

We observe that, if there are two simplices σ_i, σ_j such that for some $\rho \in \mathbb{Z}$ and $0 \leq \rho \leq \ell$ and $a, b \in \{x, y\}$, one has $|\sigma_i^a - \mathbf{p}_t^a| = \rho \cdot |\sigma_j^b - \mathbf{p}_t^b|$ then, say for $a = x$ and $b = y$, the point representing the vector $\mathbf{v}_f$ lies on a hyperplane in $\mathbb{R}^{2n}$:

$$\left\{\mathbf{v} \in \mathbb{R}^{2n} \colon\ |\mathbf{v}[2i-1] - \mathbf{p}_t^x| = \rho \cdot |\mathbf{v}[2j] - \mathbf{p}_t^y|\right\}.$$

Corresponding to each such pair of simplex coordinates, we get one hyperplane. Here, in the example, we chose one simplex coordinate to be x-coordinate and the other to be y-coordinate. This, however, need not always be the case: we can have all four possible combinations of x- and y-coordinates, corresponding to each of which we get a hyperplane. We refer the reader to Appendix A in [40] for the exact formula of all such hyperplanes. Combining all

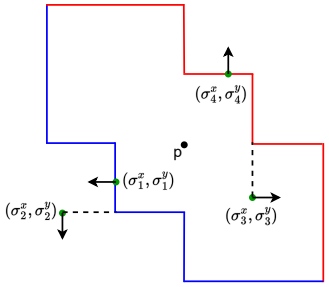


Figure 1 A 2-worm with lower boundary colored in blue and upper boundary colored in red. The figure also shows the possible cases of **constraining simplex coordinates**; σ_1 is a *lower x-constraining* simplex coordinate because the x-coordinate σ_1^x constrains the lower boundary of the worm and prevents the worm from expanding to the left; σ_2 shows a *lower y-constraining* simplex coordinate because σ_2^y also constrains the lower boundary of the worm and prevents the worm from expanding downwards. Similarly, σ_3 and σ_4 are *upper x-constraining* and *upper y-constraining* respectively. The arrows depict the gradient directions as described in Theorem 13.

these hyperplanes, we get an *arrangement $\mathcal{H}$ of hyperplanes* in $\mathbb{R}^{2n}$ [26]. The arrangement $\mathcal{H}$ partitions $\mathbb{R}^{2n}$ into *relatively open* r-cells, $r \in \{0, \ldots, 2n\}$. We observe that this arrangement induces an affine stratification $\mathcal{S}_{\mathcal{H}}$ (refer [34] for the formal definition) of $\mathbb{R}^{2n}$ where the r-dimensional strata are precisely the r-cells.

In the following we prove that $\mathcal{G}$ is a piecewise affine map relative to this arrangement, meaning that it is affine on each stratum of $\mathcal{S}_{\mathcal{H}}$. To this end, we introduce the notions of *upper boundary*, *lower boundary*, and *constraining simplex coordinate* for an ℓ-worm which will allow us to characterize the strata of $\mathcal{S}_{\mathcal{H}}$ in terms of conditions on the bifiltration function. We give an intuitive explanation of these concepts using Figure 1. We refer the reader to Appendix A in [40] for the formal definitions.

In Figure 1, a 2-worm is shown with its lower boundary colored in blue and its upper boundary in red. As depicted, the *lower boundary* consists of the part of the boundary that is in the lower half of the worm, while the *upper boundary* consists of the rest of the boundary, i.e., a point belongs to the lower boundary (resp. upper boundary) if the open lower-set (resp. upper-set) of the point does not intersect the worm.

For an intuitive understanding of *constraining simplex coordinate*, consider the Gril value d of an ℓ-worm centered at $\mathbf{p}$ for a given rank k. It follows from Definition 8 that there exists at least one simplex σ such that σ^x or σ^y prevents the worm from expanding any further to have a Gril value more than d at $\mathbf{p}$. The coordinate in question (either σ^x or σ^y) is called a *constraining simplex coordinate* for the ℓ-worm centered at $\mathbf{p}$. Here, by preventing the worm from expanding, we mean that if the worm were to expand then the value of the generalized rank over the worm would drop below the given value of rank k. Note that an ℓ-worm can have multiple constraining simplex coordinates, including some coming from the same simplex σ.

We can characterize the top-dimensional ($2n$-dimensional) strata of $\mathcal{S}_{\mathcal{H}}$ in terms of conditions on constraining simplex coordinates for ℓ-worms, and consequently, in terms of conditions on the bifiltration functions.

▶ **Proposition 10.** *The top-dimensional strata of $\mathcal{S}_{\mathcal{H}}$ consist precisely of those bifiltration functions that have a unique constraining simplex coordinate for each ℓ-worm.*

We state the main theorem of this subsection.

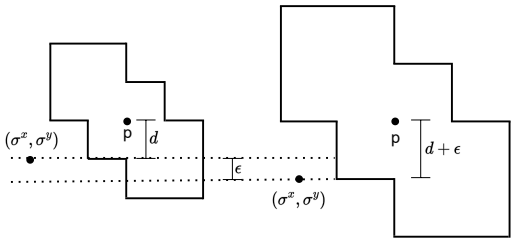


Figure 2 An intuitive understanding of the gradient assignment described in Theorem 13. The y-coordinate σ^y is at a distance of d from the y-coordinate of $\mathbf{p}$ in the left figure; σ is the only constraining simplex for the 2-worm. In the figure on the right, σ^y has *reduced* by ϵ and is now at a distance $d+\epsilon$ from $\mathbf{p}^y$. As a consequence, the value of GRIL increases from d to $d+\epsilon$. Thus, $\frac{\partial \Lambda^{\mathbf{P}}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^y} = -1$.

▶ **Theorem 11.** *Let K be a simplicial complex with n simplices. Let $k, \ell \in \mathbb{N}$, and let $\{\mathbf{p}_j\}_{j=1}^s$ be the s sampled center points for the ℓ-worms. Then, $\mathcal{G}$, as defined in Eq.*(2)*, is a piecewise affine map relative to the arrangement $\mathcal{H}$.*

Overview of the proof: $\mathcal{G}$ depends affinely on the simplex coordinates in each top-dimensional stratum, because there is a unique constraining simplex coordinate for each ℓ-worm. By continuity of $\mathcal{G}$ (Proposition 9), the restriction of $\mathcal{G}$ to each (affine) lower dimensional stratum is affine.

4.2 Stochastic sub-gradient descent

In our machine learning pipeline, the GRIL map $\mathcal{G}$ is post-composed with a loss function $N\colon \mathbb{R}^s \to \mathbb{R}$, derived e.g. from some neural network. In the corollary below, we give sufficient conditions on N that ensure the convergence of stochastic sub-gradient descent on $N \circ \mathcal{G}$.

▶ **Corollary 12.** *If $N\colon \mathbb{R}^s \to \mathbb{R}$ is definable and locally Lipschitz continuous, then, under the assumptions of Proposition 6 and Theorem 11, stochastic sub-gradient descent on $N \circ \mathcal{G}$ converges almost surely to critical points of $N \circ \mathcal{G}$.*

Overview of the proof: In the discussion towards the end of section 3.3, we saw that any piecewise affine map is definable. Since $\mathcal{G}$ is piecewise affine, it is locally Lipschitz and definable. We know that the composition of two definable functions is definable, and that the composition of two locally Lipschitz functions is locally Lipschitz. These facts, put together with Proposition 6, immediately lead to the fact that stochastic sub-gradient descent converges almost surely to critical points of $N \circ \mathcal{G}$.

4.3 Differential of GRIL

In order to back-propagate gradients through the GRIL computation, we need an explicit formula for the differential of $\mathcal{G}$. Such a formula is needed only in the top-dimensional strata, where $\mathcal{G}$ is actually smooth. In lower-dimensional strata, sub-gradients can be approximated using, e.g., gradient sampling [3].

In section 4.1, we saw that $\mathcal{G}$ is piecewise affine, which implies that its differential is constant in each top-dimensional stratum. In order to compute it, we need to determine the sign of the gradient at different simplex coordinates. To do so, given a worm, we distinguish between simplex coordinates that constrain the lower boundary of the worm and those that constrain its upper boundary. Since each simplex has two coordinates, x and y, and these can

be constraining either the upper boundary or the lower boundary, we get four different cases: *lower x-constraining, lower y-constraining, upper x-constraining* and *upper y-constraining.* All these cases are shown in Figure 1.

We deduce an explicit formula for the differential of $\mathcal{G}$ in the top-dimensional strata.

▶ **Theorem 13.** *Let K be a simplicial complex with n simplices. Let $k, \ell \in \mathbb{N}$ and $\{\mathbf{p}_j\}_{j=1}^{s}$ be the s sampled center points for the ℓ-worms. Then, the differential of $\mathcal{G}$ at any $\mathbf{v}_f$ in a top-dimensional stratum is given by:*

$$\begin{pmatrix} \frac{\partial \Lambda^{\mathbf{P}^1}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^x_1} & \frac{\partial \Lambda^{\mathbf{P}^1}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^y_1} & \frac{\partial \Lambda^{\mathbf{P}^1}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^x_2} & \cdots & \frac{\partial \Lambda^{\mathbf{P}^1}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^y_n} \\ \vdots & & & & \vdots \\ \frac{\partial \Lambda^{\mathbf{P}^s}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^x_1} & \cdots & \cdots & & \frac{\partial \Lambda^{\mathbf{P}^s}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^y_n} \end{pmatrix}_{s \times 2n}$$

where,

$$\frac{\partial \Lambda^{\mathbf{p}_j}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^x_i} = \begin{cases} -1, & \text{if } \sigma_i \text{ is lower } x\text{-constraining for } \boxed{\mathbf{p}_j}^{\ell}_{d_j}, \\ +1, & \text{if } \sigma_i \text{ is upper } x\text{-constraining for } \boxed{\mathbf{p}_j}^{\ell}_{d_j}, \\ 0, & \text{otherwise,} \end{cases}$$

$$\frac{\partial \Lambda^{\mathbf{p}_j}_{k,\ell}(\mathbf{v}_f)}{\partial \sigma^y_i} = \begin{cases} -1, & \text{if } \sigma_i \text{ is lower } y\text{-constraining for } \boxed{\mathbf{p}_j}^{\ell}_{d_j} \\ +1, & \text{if } \sigma_i \text{ is upper } y\text{-constraining for } \boxed{\mathbf{p}_j}^{\ell}_{d_j} \\ 0, & \text{otherwise,} \end{cases}$$

and d_j is the Gril *value at $\mathbf{p}_j$.*

Recall that each $\mathbf{v}_f$ in a top-dimensional stratum of $\mathcal{S}_{\mathcal{H}}$ corresponds to a bifiltration function f with a unique constraining simplex. Refer to Figure 2 for an intuitive explanation about the sign of the gradient and the different cases in Theorem 13, as shown by the arrows in Figure 1.

▶ **Corollary 14.** *Given the conditions of Theorem 13, partial derivatives of $\mathcal{G}$ with respect to $\mathbf{p}^x_j$ and $\mathbf{p}^y_j$ also exist and are given by:*

$$\frac{\partial \Lambda^{\mathbf{p}_j}_{k,\ell}(\mathbf{v}_f)}{\partial \mathbf{p}^x_j} = \begin{cases} +1, & \text{if there exists a lower } x\text{-constraining} \\ & \text{simplex for } \boxed{\mathbf{p}_j}^{\ell}_{d_j}, \\ -1, & \text{if there exists an upper } x\text{-constraining} \\ & \text{simplex for } \boxed{\mathbf{p}_j}^{\ell}_{d_j}, \\ 0, & \text{otherwise,} \end{cases}$$

$$\frac{\partial \Lambda^{\mathbf{p}_j}_{k,\ell}(\mathbf{v}_f)}{\partial \mathbf{p}^y_j} = \begin{cases} +1, & \text{if there exists a lower } y\text{-constraining} \\ & \text{simplex for } \boxed{\mathbf{p}_j}^{\ell}_{d_j}, \\ -1, & \text{if there exists an upper } y\text{-constraining} \\ & \text{simplex for } \boxed{\mathbf{p}_j}^{\ell}_{d_j}, \\ 0, & \text{otherwise.} \end{cases}$$

5 Experimental setup

In this section, we outline the overall experimental setup. We first describe some practical considerations involved in implementing D-Gril, and then describe the architecture used across all experiments.

5.1 Practical considerations

In section 4.1 we saw that $\mathcal{G}$ is piecewise affine and in section 4.2 we deduced that stochastic sub-gradient descent converges almost surely to critical points of $N \circ \mathcal{G}$ for any definable and locally Lipschitz loss function N. We gave an explicit formula for the differential of Gril in the top-dimensional strata in section 4.3, and at the beginning of that section we argued that gradient sampling can be used to approximate sub-gradients in lower-dimensional strata. In practice, for the sake of computational efficiency, we use a simple variant of gradient sampling, which consists of sampling a single nearby point and taking its gradient.

So far, we considered a bifiltration function on the entire simplicial complex K. However, in practice, we may need to extend the filtration function to the entire simplicial complex based on filtration function values on certain simplices. For example, in a lower-star filtration, the filtration function values on the higher dimensional simplices are dictated by the filtration function values on the vertices of the simplicial complex. In a Rips filtration, the filtration function values on the vertices are 0, while the filtration function values on higher dimensional simplices are dictated by the filtration function values on the edges.

In such scenarios, the bifiltration function f is given on a subcomplex L of the simplicial complex K. This is extended to a bifiltration function $\bar{f}$ on the entire simplicial complex in a piecewise constant manner. We use the example of lower-star bifiltration to show how our framework fits to this type of scenario. Let $f\colon K^0 \to \mathbb{R}^2$ be a function on the vertices of the simplicial complex K with m vertices and n simplices. The map f is extended to a piecewise constant map $\bar{f}\colon K \to \mathbb{R}^2$ as follows: for an edge $e = (u, v)$, we let $\bar{f}_x(e) = \max\{f_x(u), f_x(v)\}$ and $\bar{f}_y(e) = \max\{f_y(u), f_y(v)\}$; the values on higher dimensional simplices are defined inductively. Notice that for each simplex $\sigma \in K$, there is a *maximal x-vertex* and a *maximal y-vertex*, the vertices that give the x and y-values to the simplex respectively. We know that f can be represented as a vector $\mathbf{v}_f \in \mathbb{R}^{2m}$ and $\bar{f}$ as a vector $\mathbf{v}_{\bar{f}} \in \mathbb{R}^{2n}$ after ordering the simplices of K. Consider the map $q\colon \mathbb{R}^{2m} \to \mathbb{R}^{2n}$ given by $q(\mathbf{v}_f) = \mathbf{v}_{\bar{f}}$. The map q is piecewise affine and thus, $\mathcal{G} \circ q$ is piecewise affine as well. Therefore, we can apply stochastic sub-gradient descent on $N \circ \mathcal{G} \circ q$ with analogous convergence guarantees.

Observe that we get an arrangement of hyperplanes on $\mathbb{R}^{2m}$, $\mathcal{H}_q$, analogous to $\mathcal{H}$ on $\mathbb{R}^{2n}$ that we saw in the previous subsections. On the top-dimensional cells of $\mathcal{H}_q$, the differential of q is well-defined and constant and expressions analogous to the ones in Theorem 13 can be derived, and the chain rule can be used for back-propagating gradients. As mentioned previously, we use a simple form of gradient sampling to approximate sub-gradients when the gradient is not well-defined. For the experimental section that follows, N is a neural network with one of the standard choices for loss functions, which is definable as discussed towards the end of section 3.2.

We observe that if the bifiltration f is extended to $\bar{f}$ as described above , then we end up in a scenario where multiple simplices have the same x or y-values. This is a degenerate scenario with respect to the assumptions in Theorem 11. To avoid this, in practice, we add an infinitesimal perturbation to the higher dimensional simplices to ensure the values are not exactly the same. We note that this is, in spirit, similar to gradient sampling.

5.2 Model architecture

Every instance in each dataset is an attributed graph with a label associated with it. We use a Graph Isomorphism Network (GIN) [49], as a candidate GNN architecture, to obtain a bifiltration function over the vertices. We consider the lower-star bifiltration of this function. This gives us a bifiltration function on the entire graph, which we denote as f. Let $\mathbf{v}_f$ denote the vector representation of f which we get after ordering the n simplices of the graph. We divide the range of f into a grid of size 100×100. We randomly initialize the s center points from this grid and compute the vector $\mathcal{G}_{k,\ell}(\mathbf{v}_f)$ (Eq. (2)) for $k = 1, 2, 3$ and $\ell = 2$ at these center points. This vector $\mathcal{G}_{k,\ell}(\mathbf{v}_f)$ is fed to a 3-layer Multilayer Perceptron (MLP) classifier and the loss is back-propagated through the GRIL computation according to the gradient described in Section 4. Refer to Figure 3 for an illustration. Note that the positions of the s center points are also optimized as the gradient can be back-propagated to the coordinates of the center points as well. We refer the reader to Appendix B in [40] for additional details.

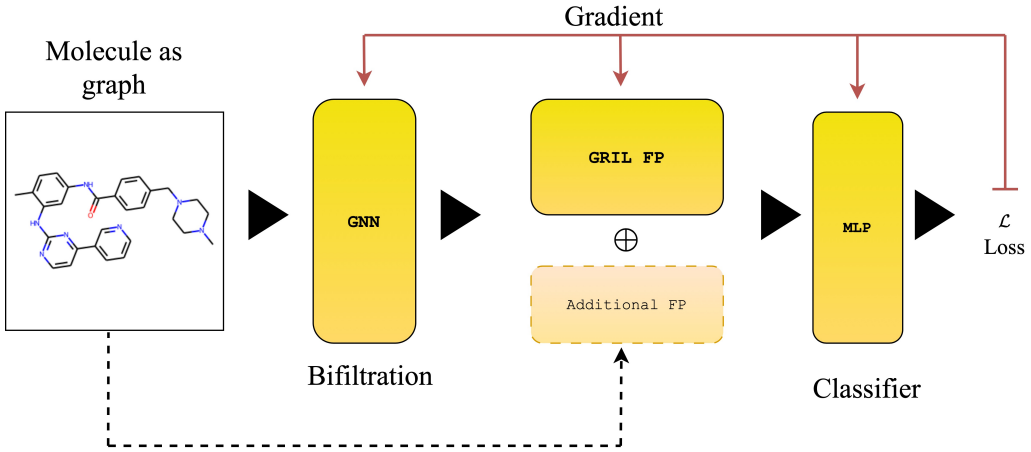


Figure 3 Architecture choice for bio-activity prediction; the bifiltration fuction f is learnt compared to the standard multiparameter pipeline; $\oplus$ denotes concatenation of vectors.

6 Experimental results

In this section, we present the experimental results on various bio-activity prediction datasets and benchmark graph datasets. We begin the section with a brief description of bio-activity prediction datasets followed by experimental results on those datasets. Then, in section 6.2, we show that D-GRIL can be, more generally, applied in the context of bifiltration learning on standard benchmark graph datasets. We compare with existing multiparameter persistent homology methods on these datasets. All the reported accuracy/ROC-AUC scores, in this section, are 5-fold cross-validated scores.

6.1 Bio-activity prediction datasets

The data is extracted from the ChEMBL database [24, 13]. Each dataset contains the SMILES [46] encoding of a novel drug (compound), and activity pairs for a target of interest. For example, the EGFR dataset contains all the molecules that have been tested against epidermal growth factor receptor (EGFR) kinase, and their measured bio-activity. The

Table 1 Test ROC-AUC on ChEMBL datasets. D-GRIL performs better than GIN with *sum* pooling and GRIL with bifiltration obtained from pre-trained GIN.

Dataset	GIN	GIN-GRIL	GIN-MPSM	D-GRIL
EGFR	55.60 ± 8.61	58.39 ± 2.51	$\mathbf{66.97 \pm 1.87}$	65.38 ± 5.34
ERBB2	57.06 ± 8.58	61.28 ± 3.66	68.75 ± 1.98	$\mathbf{69.24 \pm 5.06}$
CHEMBL1163125	58.48 ± 4.37	54.13 ± 1.09	61.23 ± 2.47	$\mathbf{65.69 \pm 1.57}$
CHEMBL2148	52.88 ± 0.91	50.24 ± 0.18	51.45 ± 0.87	$\mathbf{53.13 \pm 3.70}$
CHEMBL4005	55.90 ± 4.63	51.85 ± 3.55	$\mathbf{61.35 \pm 1.49}$	59.34 ± 5.70

SMILES encodings of the molecules are then converted to graphs with MOLFEAT [41]. Bio-activity is measured in *half maximal inhibitory concentration* (IC_{50}), which qualitatively indicates how much a drug is needed, *in vitro*, to inhibit a particular process by 50%. To facilitate the comparison of IC_{50} values, it is common practice to convert IC_{50} to $pIC_{50} = -\log_{10}(IC_{50})$, expressed in molar units. Threshold for activity cutoff $pIC_{50} = 6.3$ is used throughout the paper. We refer the reader to Appendix B in [40] for additional details.

In the first set of experiments, we compare D-GRIL with (i) a standard GNN model– GIN with sum-pooling, (ii) with GIN and GRIL as a readout layer and with (iii) GIN and MPSM [44]. Note that for the results on GIN-GRIL, GRIL is used as a passive readout layer, i.e., we obtain a bifiltration function from a pre-trained GIN (pre-trained for graph classification on the same dataset) and compute GRIL on it. We use the exact same pipeline as GIN-GRIL and train it end-to-end using D-GRIL and GIN-MPSM and report the results. The classifier, 3-layer MLP, is the same for all the experiments. Refer to Figure 3 for the pipeline. From Table 1, we can see that adding topological information after GIN has been trained can improve the performance in some cases and need not be beneficial in others. However, it is clear that adding topological information in an end-to-end framework appears to be beneficial for bio-activity prediction. D-GRIL has comparable performance to MPSM.

Further, we perform a series of experiments where we augment other popular molecular fingerprints such as ECFP and Morgan3 [43, 37] with the D-GRIL framework. In these experiments, we compare the performance of the model with and without D-GRIL augmentation. We report the results in Table 2. We see from the table that augmenting with D-GRIL seems to, generally, increase the performance of the model. Fingerprints such as ECFP and Morgan3 can essentially represent an infinite number of different molecular features, including stereochemical information [43] but they are not effective in capturing global features of molecules such as size and shape [4]. Combining D-GRIL with these fingerprints augments the model with topological information, yielding an improved performance.

6.2 Benchmark graph datasets

D-GRIL can be used more generally for filtration learning on graph datasets. We perform a series of experiments with benchmark graph datasets such as MUTAG, PROTEINS, DHFR, COX2 [38] and compare with existing multiparameter persistence methods. Details about the datasets are given in Appendix B in [40]. We report the results in Table 3. For a valid comparison, we use a 3-layer MLP as the classifier for all the multiparameter signatures. Hence, the numbers reported look different from the ones in [48], [6] where the authors consider XGBoost [9]. In fact, in [48], the authors show that XGBoost performs much better than a 3-layer MLP classifier for GRIL vectors on these datasets. We can see from the table that learning the bifiltration function seems to perform better than multiparameter

Table 2 Test ROC-AUC scores on ChEMBL datasets; augmenting ECFP and Morgan3 fingerprints with D-GRIL increases the classification performance for most of the datasets.

Dataset	ECFP	ECFP+D-GRIL	Morgan3	Morgan3+D-GRIL
EGFR	83.27 ± 1.10	$\mathbf{84.37 \pm 1.22}$	82.39 ± 1.35	$\mathbf{83.57 \pm 1.40}$
ERBB2	83.53 ± 1.31	$\mathbf{86.24 \pm 2.27}$	83.19 ± 1.26	$\mathbf{85.51 \pm 1.47}$
CHEMBL1163125	83.94 ± 1.23	$\mathbf{86.33 \pm 0.67}$	83.55 ± 1.20	$\mathbf{84.85 \pm 0.82}$
CHEMBL203	81.74 ± 0.90	$\mathbf{81.86 \pm 0.90}$	80.85 ± 1.32	$\mathbf{81.64 \pm 0.45}$
CHEMBL2148	73.96 ± 2.59	$\mathbf{74.13 \pm 2.33}$	72.79 ± 1.97	$\mathbf{74.50 \pm 2.61}$
CHEMBL279	76.72 ± 1.34	$\mathbf{78.75 \pm 0.77}$	76.76 ± 0.50	$\mathbf{78.05 \pm 1.18}$
CHEMBL2815	73.69 ± 1.53	$\mathbf{76.51 \pm 1.48}$	73.42 ± 0.56	$\mathbf{75.22 \pm 2.85}$
CHEMBL4005	$\mathbf{80.45 \pm 1.30}$	80.41 ± 0.97	79.95 ± 1.30	$\mathbf{80.55 \pm 0.82}$
CHEMBL4722	78.05 ± 1.64	$\mathbf{78.51 \pm 1.83}$	78.76 ± 1.37	$\mathbf{79.17 \pm 1.30}$

Table 3 Test accuracies of D-GRIL on benchmark graph datasets.

Dataset	D-GRIL	MPSM	GRIL	MP-I	MP-L	MP-K	P
MUTAG	$\mathbf{85.09 \pm 5.99}$	78.44 ± 3.33	83.49 ± 3.64	74.99 ± 2.79	82.42 ± 3.72	79.27 ± 2.45	66.50 ± 0.87
PROTEINS	69.45 ± 4.11	68.37 ± 3.31	66.31 ± 2.34	70.80 ± 3.09	$\mathbf{70.80 \pm 1.31}$	61.70 ± 2.98	59.57 ± 0.08
DHFR	61.24 ± 4.37	60.97 ± 1.98	$\mathbf{61.64 \pm 1.66}$	60.98 ± 0.10	61.11 ± 0.25	60.98 ± 0.10	60.98 ± 0.10
COX2	78.16 ± 0.41	77.73 ± 0.36	78.16 ± 0.41	78.16 ± 0.41	78.16 ± 0.41	78.16 ± 0.41	78.16 ± 0.41
IMDB-BINARY	$\mathbf{62.60 \pm 6.56}$	60.80 ± 3.82	50.00 ± 0.00	56.60 ± 2.94	50.00 ± 0.00	50.30 ± 1.12	50.00 ± 0.00

Table 4 Reported times (hh:mm:ss) are training times per fold averaged over 5 folds that we used for training. All the experiments have been performed on a machine with AMD EPYC 7313 16-Core Processor and NVIDIA A10 GPU.

Dataset	D-GRIL	MPSM
MUTAG	00:05:08	00:08:47
COX2	00:11:34	00:18:32
DHFR	00:16:48	00:27:12
PROTEINS	00:30:03	00:50:26

persistence methods on popular choices of bifiltration functions on most datasets. We can also see that D-GRIL performs better than GRIL, supporting our argument for an end-to-end learning framework. We also see that D-GRIL performs better than MPSM. The training times for the two end-to-end learning methods, D-GRIL and MPSM are reported in Table 4. We can see that the training times for D-GRIL are improved by about 40% compared to the ones of MPSM. We refer the reader to Figure 4 [40] for visualizations of bifiltration functions learnt by D-GRIL.

Additionally, we performed experiments where we added a fusion layer to combine the representation power of both D-GRIL and MPSM. We present the results in Table 5. We see from the table that combining the representations leads to a better overall performance, which includes a generally higher mean and a lower standard deviation. Thus, D-GRIL can also be used with existing vectorization methods in a complementary manner to capture additional topological information.

Table 5 Test accuracies of D-Gril+Mpsm on benchmark graph datasets.

Dataset	D-GRIL	MPSM	D-GRIL + MPSM
MUTAG	85.09 ± 5.99	78.44 ± 3.33	83.68 ± 3.06
PROTEINS	69.45 ± 4.11	68.37 ± 3.31	70.23 ± 3.56
DHFR	61.24 ± 4.37	60.97 ± 1.98	62.83 ± 3.68
COX2	78.16 ± 0.41	77.73 ± 0.36	78.72 ± 0.38
IMDB-BINARY	62.60 ± 6.56	60.80 ± 3.82	64.70 ± 3.96

7 Conclusions

In this paper, we propose a differentiable framework using Gril. We show that stochastic sub-gradient descent converges almost surely on Gril, and we compute an explicit formula for the differential of Gril. We use this formula to back-propagate gradients through the Gril computation, yielding the differentiable framework D-Gril. We use D-Gril to show that adding topological information in an end-to-end manner is beneficial for bio-activity prediction. Further, we show that D-Gril can be used in a more general setting, for filtration learning on graphs. Moreover, we show that D-Gril is faster and lighter, both in theory and in practice, than Mpsm. Our results indicate that learnt bifiltration functions on graphs, generally, give a better performance than the standard choices for bifiltration functions. Recently, the Gril framework has been extended to a variant of 2-parameter persistence called *quasi-zigzag* persistence [19]. The results in this paper may shed light on the question of differentiability in the quasi-zigzag framework.

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